Chapter 13: Brain Metabolism and Blood Supply — Mathematical Equations

Comprehensive Neuroscience Reference

Abstract

This chapter presents the quantitative framework for understanding brain energy metabolism and cerebral blood flow regulation. We derive models for ATP consumption and yield (30–32 ATP per glucose via oxidative phosphorylation), glucose transport across the BBB and neuronal uptake (Michaelis-Menten kinetics with substrate-specific transporters), astrocyte glycolysis and the astrocyte-neuron lactate shuttle (ANLS) linking activity to lactate export, alternative fuel utilization (ketones yielding 29 ATP, lactate yielding 15 ATP), cerebral blood flow regulation (CO_2 sensitivity 4% CBF change per mmHg $PaCO_2$; myogenic autoregulation 60–150 mmHg MAP; pericyte-mediated neurovascular coupling), and vulnerability to metabolic disruption (ischemic cascade with core/penumbra distinction, ATP depletion timelines, excitotoxicity). Five case studies with numerical solutions demonstrate: BBB glucose transport during hypoglycemia, ATP yield comparison between glucose, lactate, and ketones, cerebral autoregulation pressure-flow relationships, neurovascular coupling BOLD response dynamics, and ischemic stroke progression with reperfusion timing. All equations include typical parameter values grounded in experimental neuroscience and clinical neurology.

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1. Brain Energy Budget and ATP Demand

1.1 Brain ATP Consumption Breakdown

The brain consumes ATP at a rate proportional to its mass and metabolic intensity:

$$\mathrm{ATP}_{\mathrm{total}} = \mathrm{ATP}_{\mathrm{Na^+/K^+-ATPase}} + \mathrm{ATP}_{\mathrm{synaptic}} + \mathrm{ATP}_{\mathrm{other}} \tag{1}$$

with approximate proportions:

$$\mathrm{ATP}_{\mathrm{Na^+/K^+-ATPase}} \approx 0.50 \cdot \mathrm{ATP}_{\mathrm{total}} \tag{2}$$

$$\mathrm{ATP}_{\mathrm{synaptic}} \approx 0.25 \cdot \mathrm{ATP}_{\mathrm{total}} \tag{3}$$

The total brain ATP consumption rate:

$$\text{ATP consumption rate} = \frac{P_{\mathrm{brain}}}{E_{\mathrm{ATP}}} \tag{4}$$

Symbol	**Meaning**	**Typical Value**
P_{brain}	Brain metabolic power	∼20 watts (1,400 kcal/day)
E_{ATP}	Energy per ATP hydrolysis	30.5 kJ/mol
$\mathrm{ATP}_{\mathrm{total}}$	Total brain ATP consumption	∼90 mmol/day or ∼6 mmol/min
$\mathrm{ATP}_{\mathrm{Na^+/K^+-ATPase}}$	ATP for maintaining ion gradients	∼45 mmol/day

Explanation: The Na^+/K^+-ATPase (Na^+/K^+-pump) is the single largest ATP consumer in the brain (∼50% of all ATP), reflective of the enormous cost of maintaining transmembrane ion gradients. Each neuron fires hundreds of action potentials per second, allowing Na^+ influx and K^+ efflux that must be reversed. Synaptic transmission accounts for ∼25%.

1.2 ATP Yield from Glucose Oxidation

Complete glucose oxidation generates approximately 30–32 ATP per glucose molecule:

$$\mathrm{Glucose} + 6\,\mathrm{O_2} + n\,\mathrm{ADP} + n\,\mathrm{P}_i \rightarrow 6\,\mathrm{CO_2} + 6\,\mathrm{H_2O} + n\,\mathrm{ATP} \tag{5}$$

where $n \approx 30$–32.

The breakdown is:

- **Glycolysis**: 2 ATP (net) + 2 NADH → ∼5 ATP equivalent
- **Pyruvate dehydrogenase**: 2 pyruvate to 2 acetyl-CoA (2 NADH → ∼5 ATP)
- **TCA cycle**: 2 turns yield 6 NADH (∼15 ATP) + 2 $FADH_2$ (∼3 ATP) + 2 GTP/ATP = ∼20 ATP
- **Total**: ∼2 + 5 + 5 + 20 = **32 ATP/glucose**

Symbol	Meaning	Typical Value
n	ATP yield per glucose	30–32 ATP
$\Delta G^{\circ\prime}$	Standard free energy of glucose oxidation	−2,870 kJ/mol
Efficiency	Energy captured as ATP / total energy	∼33%

Explanation: Under basal brain metabolism (∼5.6 mg glucose/100g/min), the brain oxidizes approximately 100 g glucose per day. This yields ∼3,200 mmol ATP per day, sustaining the 90 mmol ATP per day needed by the brain.

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2. Glucose Transport and Cellular Metabolism

2.1 Endothelial GLUT1 and Neuronal GLUT3 Transport

Glucose enters the brain via GLUT1 and is taken up by neurons via GLUT3, both following Michaelis-Menten kinetics:

$$v_{\text{GLUT1}} = \frac{V_{\text{max}}^{\text{GLUT1}} \cdot [\text{Glucose}]_{\text{blood}}}{K_m^{\text{GLUT1}} + [\text{Glucose}]_{\text{blood}}} \tag{6}$$

$$v_{\text{GLUT3}} = \frac{V_{\text{max}}^{\text{GLUT3}} \cdot [\text{Glucose}]_{\text{astrocyte}}}{K_m^{\text{GLUT3}} + [\text{Glucose}]_{\text{astrocyte}}} \tag{7}$$

Symbol	Meaning	Typical Value
$V_{\text{max}}^{\text{GLUT1}}$	Maximum GLUT1 transport rate	∼100–150 nmol/(cm^2·min)
$V_{\text{max}}^{\text{GLUT3}}$	Maximum GLUT3 transport rate	∼50–100 nmol/(min·mg protein)

Symbol	Meaning	Typical Value
K_m^{GLUT1}	Michaelis constant for GLUT1	∼1.7 mmol/L
K_m^{GLUT3}	Michaelis constant for GLUT3	∼0.5–1.0 Glucose mmol/L Glucose>
$_{\text{blood}}$	Blood glucose concentration	5 mmol/LGlucose (fasting) Glucose>
$_{\text{astrocyte}}$	Astrocytic glucose concentration	∼1–2 mmol/L

Explanation: GLUT1 has higher Km than GLUT3, meaning GLUT1 is more efficient at low glucose concentrations, ensuring steady BBB glucose supply. GLUT3's lower Km means neurons preferentially take up glucose when astrocyte glucose is available.

2.2 Astrocyte Glycolysis and Pyruvate Production

Astrocytes are more glycolytic than neurons and constitute the primary site of glucose metabolism. The rate of glycolysis:

$$v_{\text{glycolysis}} = k_1 \cdot [\text{Glucose}]_{\text{astrocyte}} \cdot f(\text{activity}) \tag{8}$$

where $f(\text{activity})$ depends on intracellular calcium:

$$f(\text{activity}) = 1 + \frac{([\text{Ca}^{2+}]_{\text{in}} - [\text{Ca}^{2+}]_{\text{baseline}})}{K_{\text{Ca}}} \tag{9}$$

Symbol	Meaning	Typical Value
$v_{\text{glycolysis}}$	Glycolytic flux in astrocytes	∼0.5–2.0 μmol/(min·mg protein)
k_1	Rate constant for glucose consumption	∼0.01–0.1 Ca^{2+} min^{-1} Ca^{2+}>
$_{\text{in}}$	Intracellular astrocytic Ca^{2+} during activity	∼0.5–1.0 Ca^{2+} μM Ca^{2+}>
$_{\text{baseline}}$	Baseline astrocytic Ca^{2+} (resting)	∼0.1 μM

Symbol	Meaning	Typical Value
K_{Ca}	Sensitivity constant for Ca^{2+}-mediated activation	~0.2 μM
$f(\text{activity})$	Activity-dependent stimulation factor	1–5×

Explanation: Glutamatergic synaptic activity → astrocytic glutamate uptake → astrocytic Ca^{2+} influx → increased glycolytic enzyme activity. This activity-dependent glycolysis is the foundation of the ANLS.

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3. Astrocyte-Neuron Lactate Shuttle (ANLS)

3.1 Lactate Production and Export

Lactate production depends on glycolytic flux:

$$v_{\text{lactate, prod}} = 2 \cdot v_{\text{glycolysis}} \tag{10}$$

Lactate export follows carrier kinetics:

$$v_{\text{lactate, export}} = \frac{V_{\text{max}}^{\text{MCT1}} \cdot [\text{Lactate}]_{\text{astrocyte}}}{K_m^{\text{MCT1}} + [\text{Lactate}]_{\text{astrocyte}}} \tag{11}$$

Symbol	Meaning	Typical Value
$v_{\text{lactate, prod}}$	Lactate production rate in astrocytes	~1.0–4.0 μmol/(min·mg)
$v_{\text{lactate, export}}$	Lactate export rate via MCT1	~0.5–2.0 μmol/(min·mg)
$V_{\text{max}}^{\text{MCT1}}$	Maximum MCT1 transport rate	~200–500 nmol/(cm^2·min)
K_m^{MCT1}	Michaelis constant for MCT1	~4–5 mmol/L
Lactate$_{\text{astrocyte}}$	Intracellular astrocytic lactate	Lactate > ~2–5 mmol/L during activity

Explanation: During active neural transmission, astrocytes produce lactate faster than it can be oxidized locally, causing lactate to accumulate and be exported. This lactate is taken up by nearby neurons via MCT2.

3.2 Neuronal Lactate Uptake and Oxidation

Neurons import lactate via MCT2 and convert it to pyruvate, then acetyl-CoA:

$$\text{Lactate} \rightarrow \text{Pyruvate} \rightarrow \text{Acetyl-CoA} \rightarrow \text{TCA cycle} \rightarrow 15\,\text{ATP} \tag{12}$$

The rate of neuronal lactate uptake:

$$v_{\text{lactate, neuron}} = \frac{V_{\text{max}}^{\text{MCT2}} \cdot [\text{Lactate}]_{\text{extracellular}}}{K_m^{\text{MCT2}} + [\text{Lactate}]_{\text{extracellular}}} \tag{13}$$

Symbol	Meaning	Typical Value
$V_{\text{max}}^{\text{MCT2}}$	Maximum MCT2 transport rate	∼50–100 nmol/(cm^2·min)
K_m^{MCT2}	Michaelis constant for MCT2	∼0.5–1.0 mmol/L
extracellular	Brain interstitial lactate	Lactate> ∼1–2 mmol/L (resting); 3–5 mmol/L (activity)
ATP yield per lactate	Energy from complete oxidation	∼15 ATP

Explanation: Lactate is a more efficient fuel than glucose for direct neuronal energy production because it bypasses glycolysis, yielding 15 ATP per lactate.

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4. Alternative Brain Fuels

4.1 Ketone Body Metabolism

Ketones cross the BBB via MCT1 transporters and are metabolized:

$$\beta\text{-hydroxybutyrate} \rightarrow \text{Acetoacetate} \rightarrow 2\,\text{Acetyl-CoA} \rightarrow \text{TCA cycle} \rightarrow 29\,\text{ATP} \tag{14}$$

The rate of ketone utilization:

$$v_{\text{ketone}} = \frac{V_{\text{max}}^{\text{ketone}} \cdot [\text{Ketones}]_{\text{blood}}}{K_m^{\text{ketone}} + [\text{Ketones}]_{\text{blood}}} \tag{15}$$

Symbol	Meaning	Typical Value
$[\text{Ketones}]_{\text{blood}}$	Plasma ketone concentration	0.1–0.2 (fed); 1–5 (fasting); 5–10 (ketogenic diet) mmol/L
K_m^{ketone}	Michaelis constant for ketone transporters	∼1–2 mmol/L
$V_{\text{max}}^{\text{ketone}}$	Maximum ketone utilization rate	∼50–100 μmol/(min·100g)
ATP yield per ketone pair	Energy from β-hydroxybutyrate	∼29 ATP

Explanation: Ketones are more oxygen-efficient than glucose — they yield 29 ATP per molecule but produce less ROS (reactive oxygen species) during oxidation, making them neuroprotective under hypoxic/ischemic conditions. The brain can derive up to 60–70% of its ATP from ketones during prolonged fasting.

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5. Cerebral Blood Flow Regulation

5.1 CO_2-Mediated Vasodilation

Carbon dioxide is the most potent cerebral vasodilator:

$$\Delta CBF = \text{CBF}_{\text{baseline}} \times k_{\text{CO}_2} \times (\Delta P_{aCO_2}) \tag{16}$$

Symbol	Meaning	Typical Value
$\text{CBF}_{\text{baseline}}$	Baseline cerebral blood flow	∼50 mL/(min·100g)
k_{CO_2}	CO_2 sensitivity coefficient	∼4% CBF change per mmHg $PaCO_2$

Symbol	Meaning	Typical Value
ΔP_{aCO_2}	Change in arterial CO_2 partial pressure	±5 mmHg
P_{aCO_2}	Arterial CO_2 partial pressure	35–45 mmHg (normal; ∼40 mmHg)

Example:

- Hyperventilation → $PaCO_2$ drops from 40 to 35 mmHg (−5 mmHg) → CBF drops ∼20% → CBF ≈ 40 mL/(min·100g)
- Hypercapnia → $PaCO_2$ rises to 50 mmHg (+10 mmHg) → CBF increases ∼40% → CBF ≈ 70 mL/(min·100g)

Explanation: CO_2/H^+ acts on arteriolar smooth muscle directly (acidosis → relaxation) and on astrocytes. This mechanism is clinically exploited: hyperventilation temporarily reduces intracranial pressure, while hypercapnia increases it.

5.2 Myogenic Autoregulation

The brain maintains relatively constant blood flow across a wide range of mean arterial pressures:

$$\text{CBF} = \text{CBF}_{\text{ref}} \quad \text{for} \quad 60 < \text{MAP} < 150 \text{ mmHg} \tag{17}$$

The autoregulatory efficiency:

$$\text{Autoregulatory index} = \frac{d\log(\text{CBF})}{d\log(\text{MAP})} \tag{18}$$

Symbol	Meaning	Typical Value
CBF_{ref}	Reference cerebral blood flow (maintained constant)	∼50 mL/(min·100g)
MAP	Mean arterial pressure	60–150 mmHg (autoregulatory range)

Symbol	Meaning	Typical Value
Autoregulatory index	Log-log slope (0 = perfect autoregulation)	Near 0 (healthy brain)

Explanation: Outside the 60–150 mmHg range, autoregulation fails. Below 60 mmHg, cerebral ischemia begins. Above 150 mmHg, breakthrough autoregulation leads to cerebral edema and hypertensive encephalopathy.

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6. Case Study: Neurovascular Coupling and BOLD Response

Scenario: Model the BOLD signal response to a 30-second visual stimulus, showing how neural activity drives hemodynamic changes.

Solution:

Neural activity increases baseline firing rate from ∼5 Hz to ∼10–15 Hz in visual cortex neurons.

Glutamate release increases ∼3-fold, driving astrocytic Ca^{2+} rise and glycolysis activation.

Astrocytic lactate export increases from baseline 0.5 μmol/(min·mg) to 2 μmol/(min·mg).

NO and prostaglandin production from neurons and astrocytes drives arteriolar dilation.

CBF response follows a characteristic time course: - **Onset latency**: ∼1–2 seconds (slight initial dip as oxygen is consumed) - **Peak**: ∼4–6 seconds after stimulus onset - **Magnitude**: ∼20–40% increase in local CBF (from 50 → 60–70 mL/(min·100g)) - **Undershoot**: After stimulus offset, CBF transiently falls below baseline (∼1–3% undershoot)

BOLD signal increases ∼1–2% with these hemodynamic changes (reflecting decreased deoxyhemoglobin).

Time-course equation (gamma function approximation):

$$\text{BOLD}(t) = A \cdot t^n \cdot e^{-\lambda t}$$

where A = amplitude (1–2%), $n \approx 6$ (shape parameter), $\lambda \approx 0.1$ s^{-1} (decay rate).

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